

Exit Tutor

FOR

MIS

Theme 3 :	Cr. H	Weight of course or proportion	Competencies/ General objective	Contents/ Chapters	Number of test items from Chr	Learning Outcome
MIS	3	4%	Capable of using information communication on technology in utilizing and managing organizational resources	CH 1: Introduction to MIS	1	- Understand and recognize the relationship between information and decision making, - Know significance of information and information systems as basic resources from managerial perspective in decision -making. - Appreciate the ability and skills to identify their information needs, source and to utilize the information efficiently and effectively
				CH 4: Common Business Applications of Information Technology	1	
				CH 5: Knowledge Management (KM)	1	
				CH 6: Ethical and Security Issues	1	

Introduction to MIS

- Which one is true regards to MIS definition?
 - A. MIS is defined as a system which provides information support for decision making in the organization.
 - B. MIS is defined as an integrated system of man and machine for providing the information to support the operations.
 - C. MIS is defined as a system based on the database of the organization evolved for the purpose of providing information to the people in the organization.
 - D. MIS is defined as a Computer - based Information System.
 - E. MIS is a system to support the decision making function in the organization.

Introduction to MIS

This shows what.....?



Introduction to MIS

➤ This is the Models of _____?????



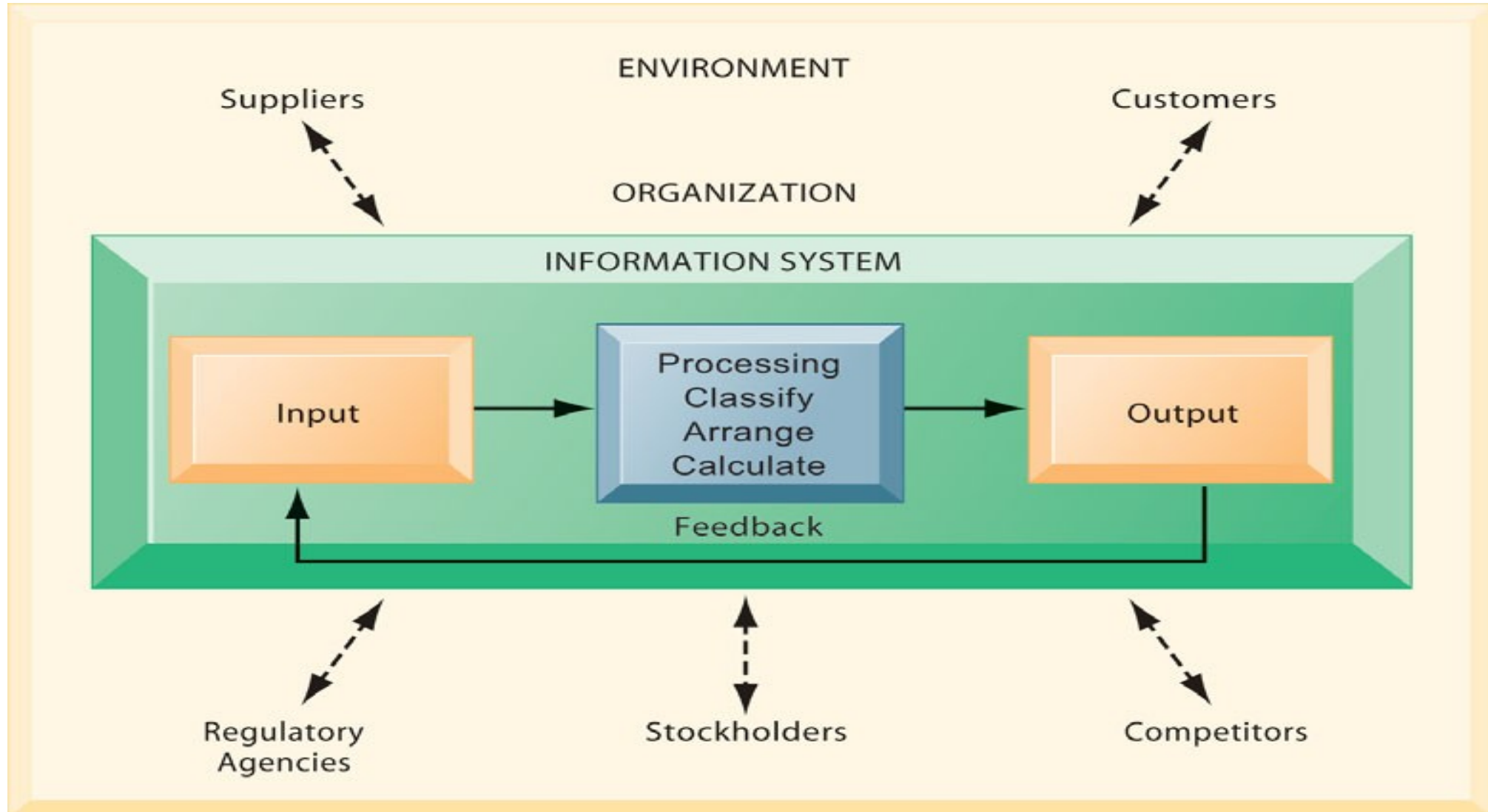
Introduction to MIS

- ❖ In today's world MIS is computerized business processing system generating information for the people in the organization to meet the information needs decision making to achieve the corporate objective of the organization.



- ❖ In any organization, small or big, **a major portion of the time goes in data collection**, processing, documenting it to the people.

Fig. Functions of an information system



Introduction to MIS

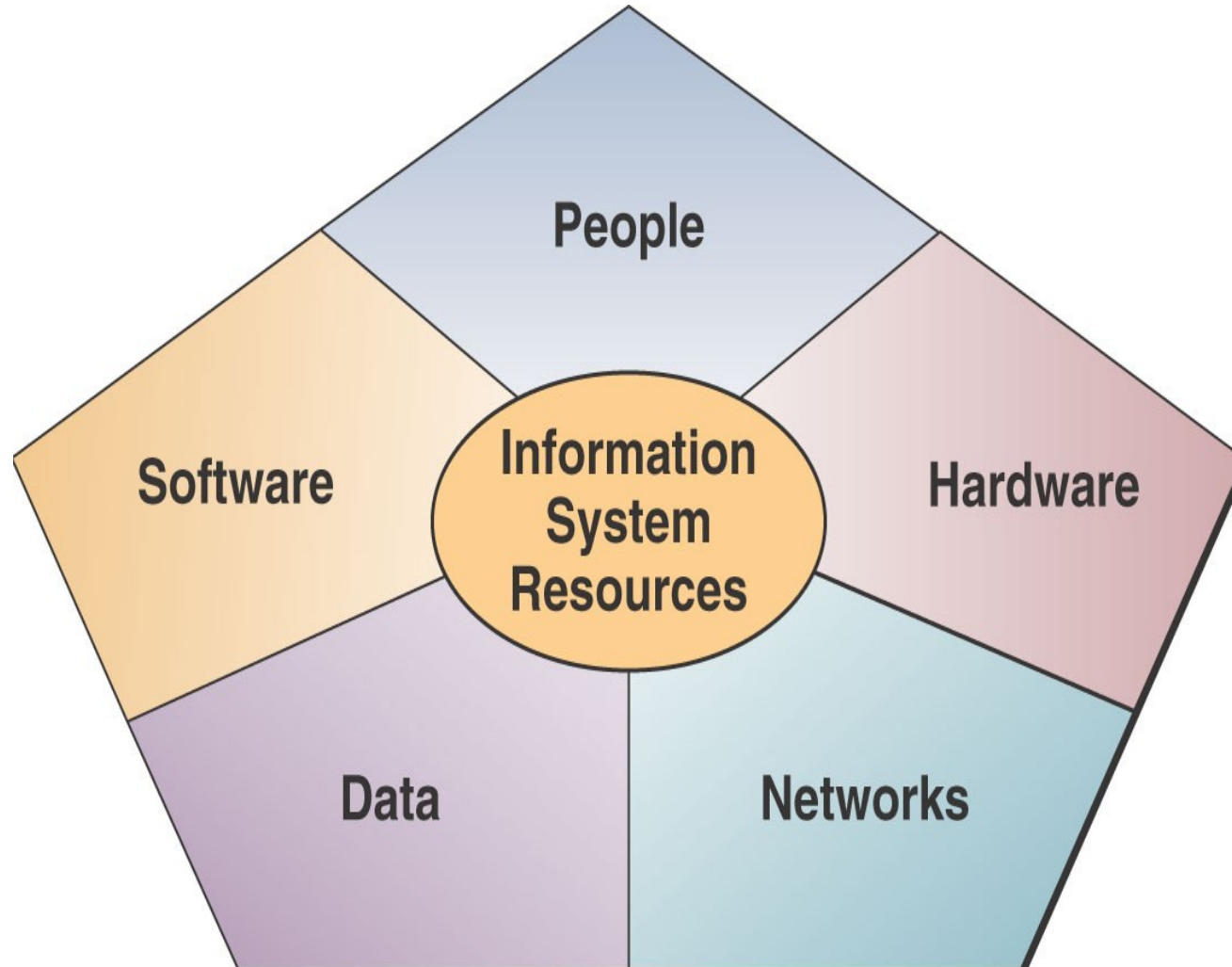


Figure: Information System Resources

NB. In one department we expect all this resource

Introduction to MIS

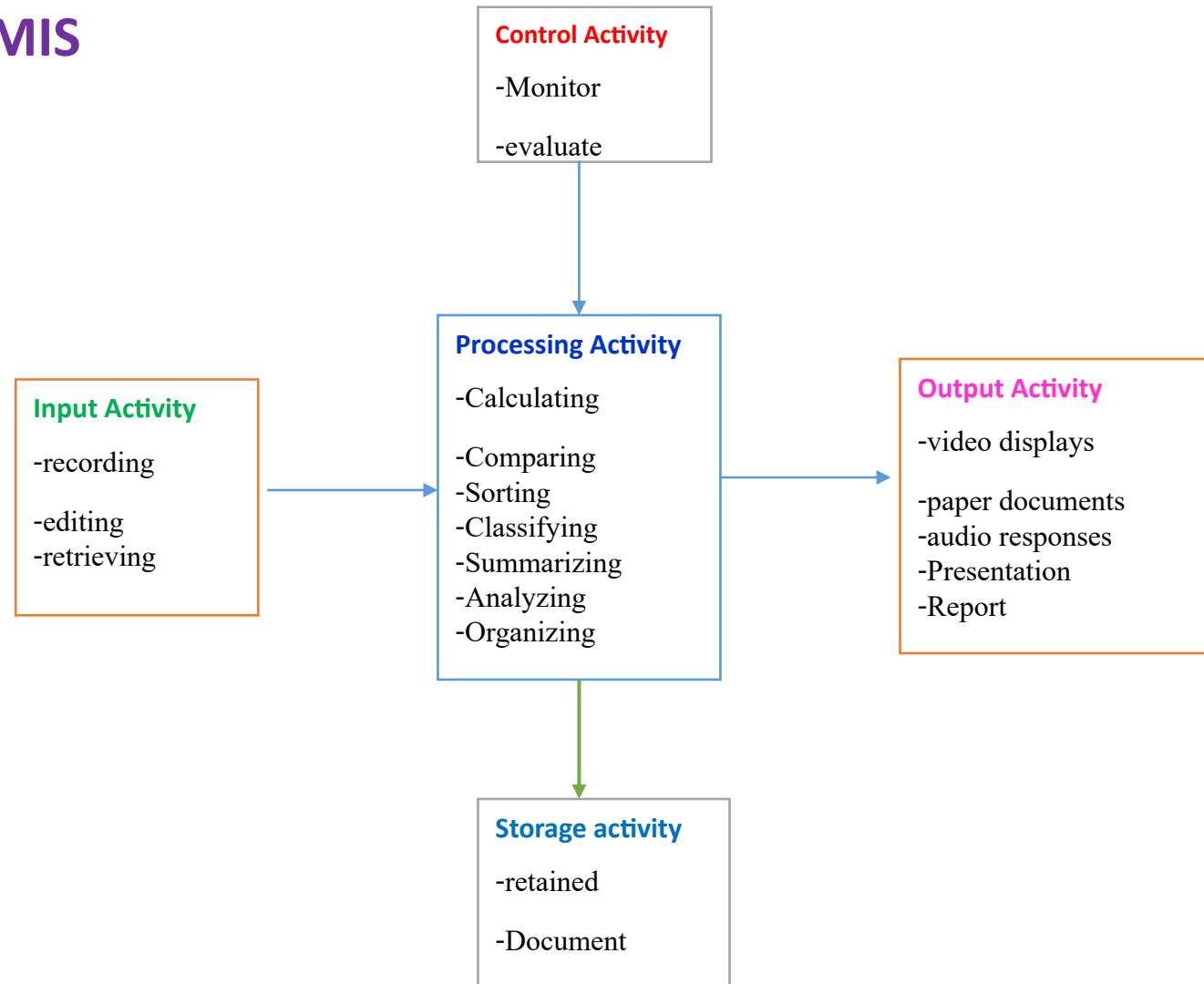
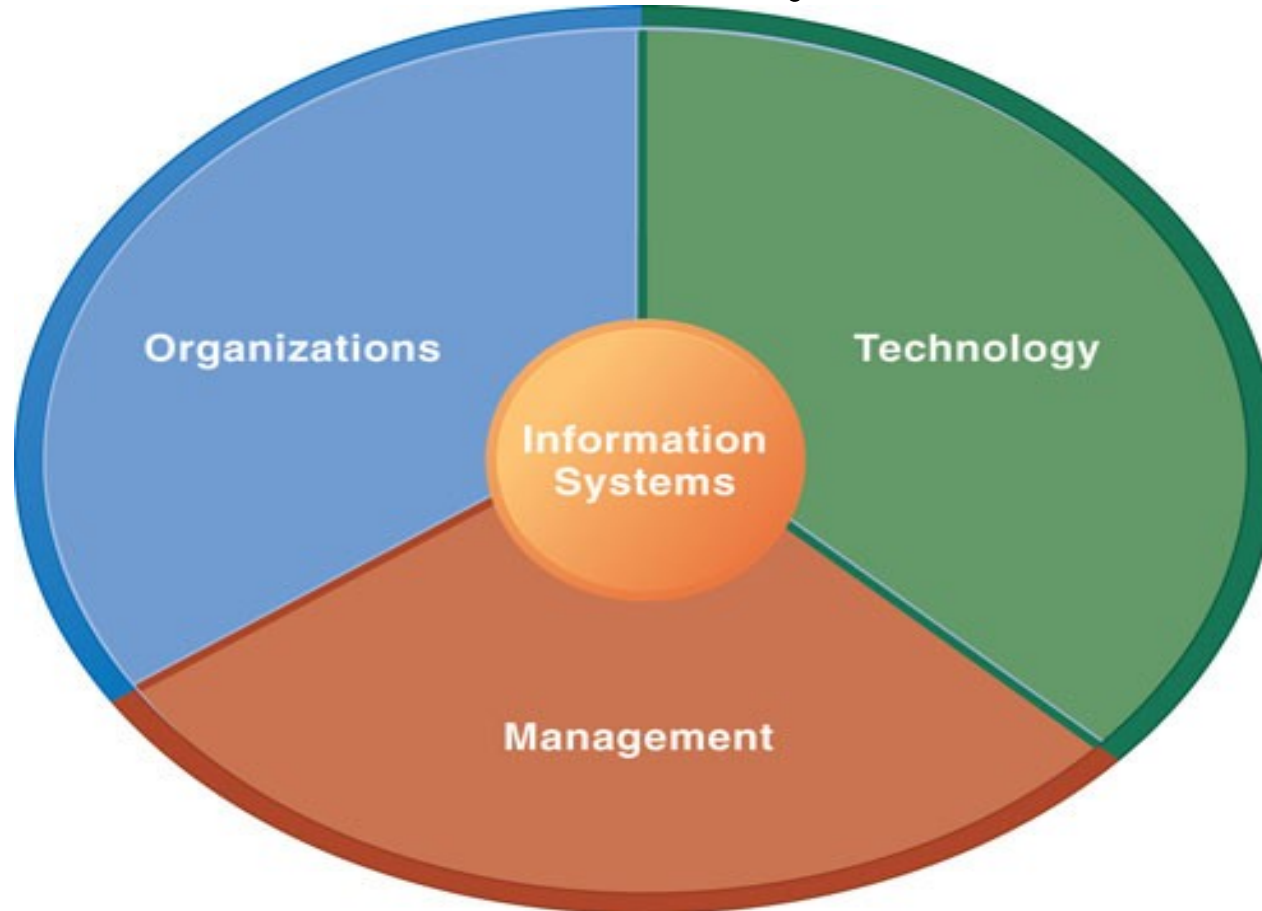


Figure 1.1. Activities of Information systems

NB. In one department , we expect all this activities

Dimensions of information systems

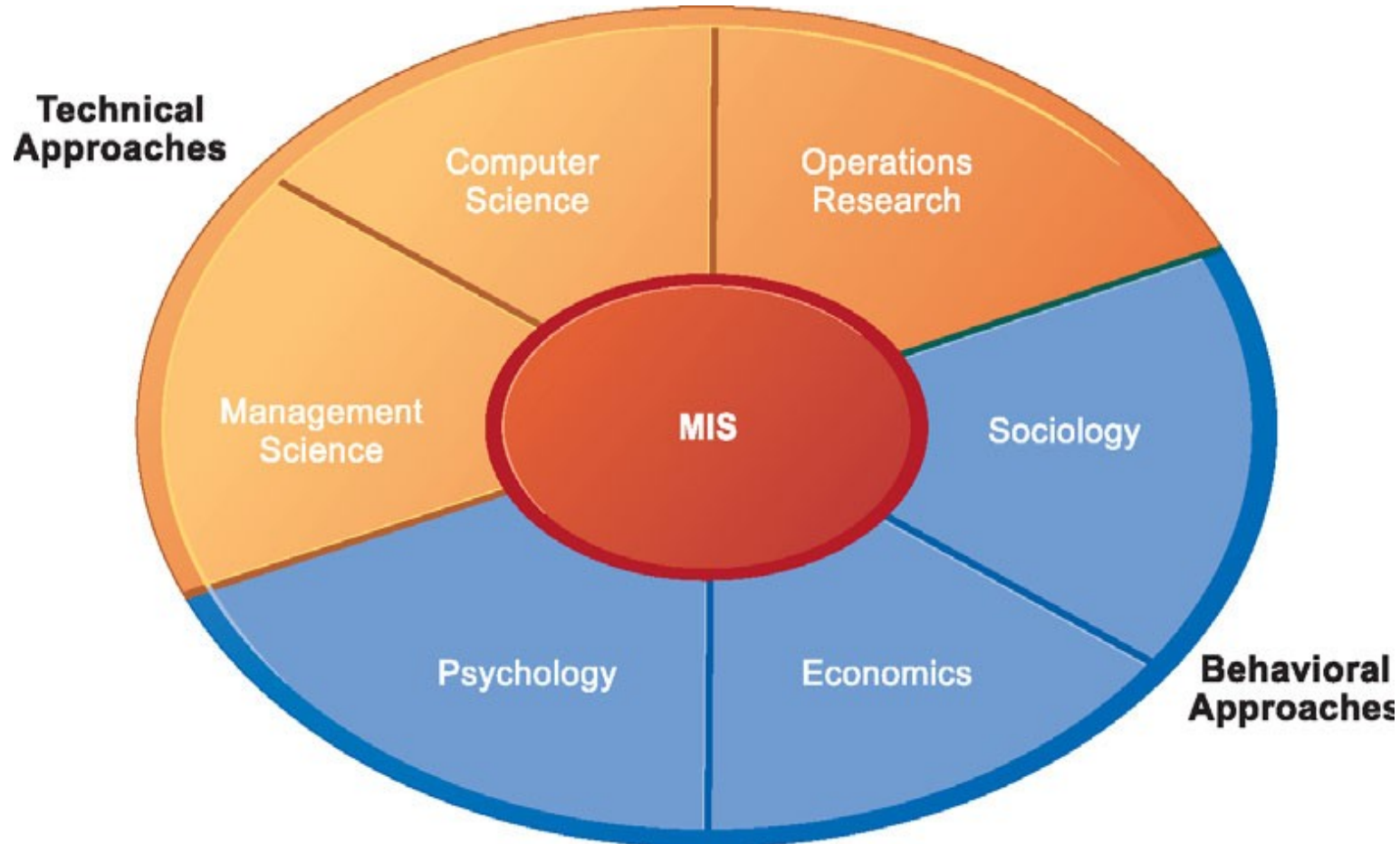


NB. So, Information systems are more than computer

Introduction to MIS

- The key elements of an **organization** are its people, structure, business processes, politics, and culture.
- **Management's job** is to make sense out of the many situations faced by organizations, make decisions, and formulate action plans to solve organizational problems.
- **Information technology** can play a powerful role in helping managers design and deliver new products and services and redirecting and redesigning their organizations. (Tools)

Contemporary approach to information system



- The study of information systems is a **multidisciplinary** field.
- **No single theory or perspective dominates**

NB, Information systems are socio-technical systems

Understand and recognize the relationship between information and decision making,

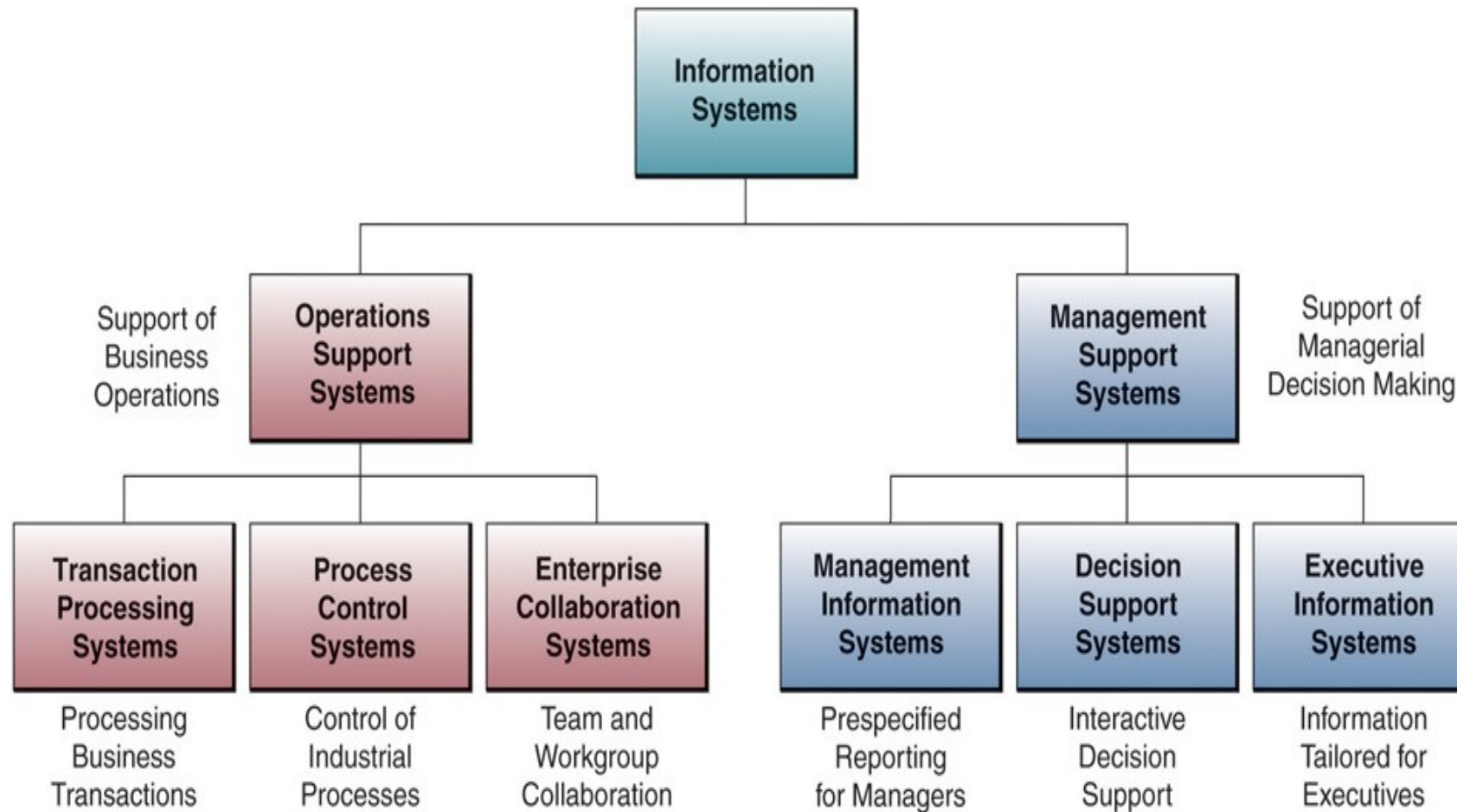
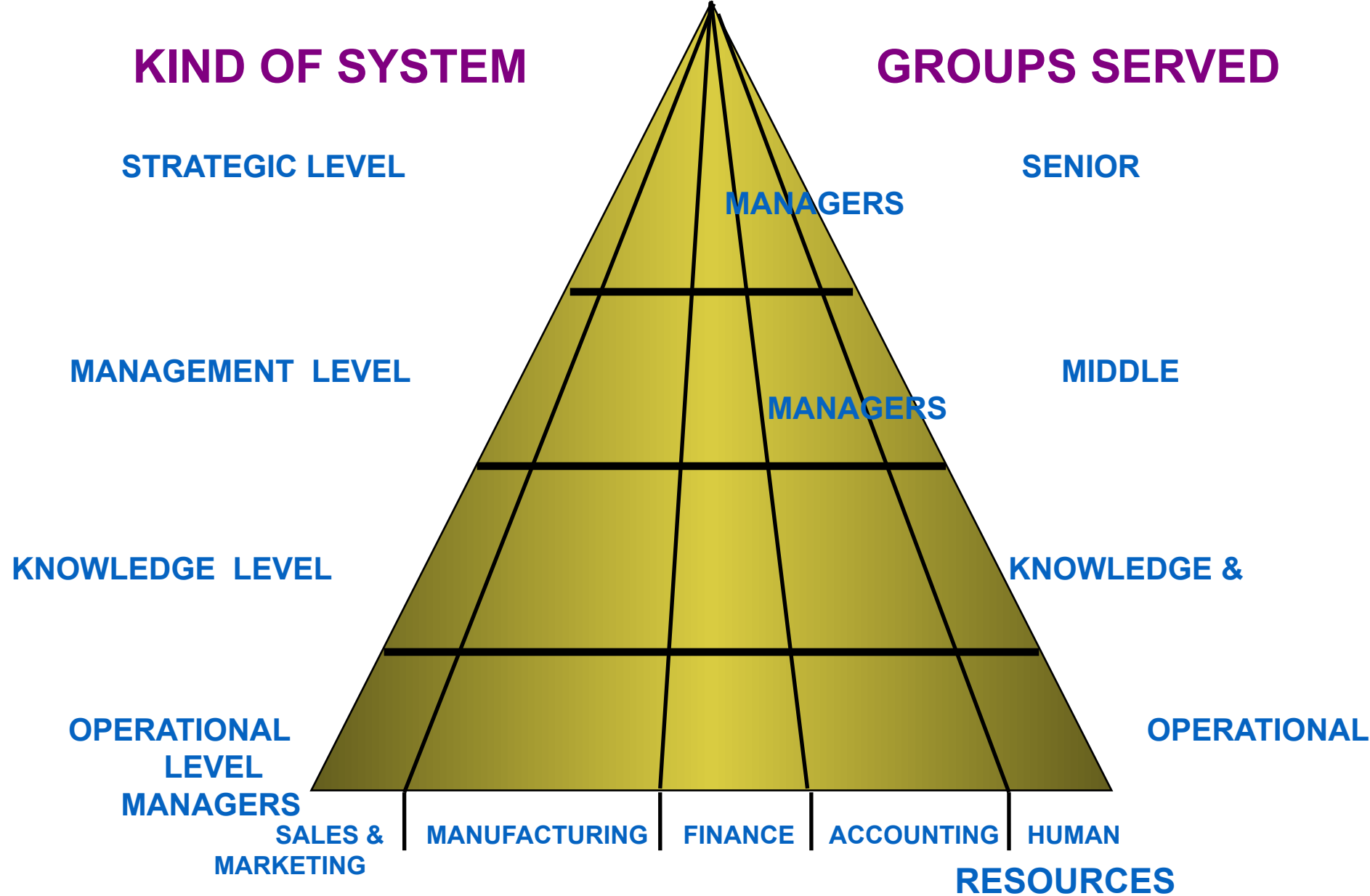
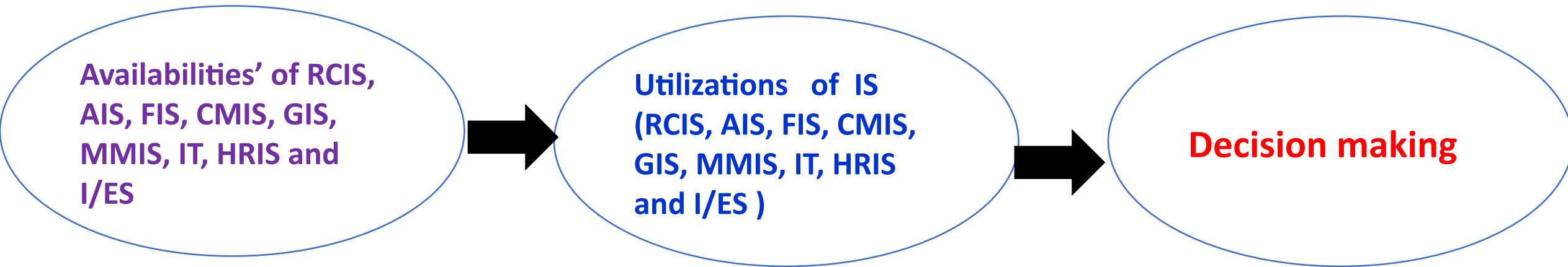


Fig1.4 Types of Information Systems

Know significance of **information and information systems** as basic resources from managerial perspective in decision-making.



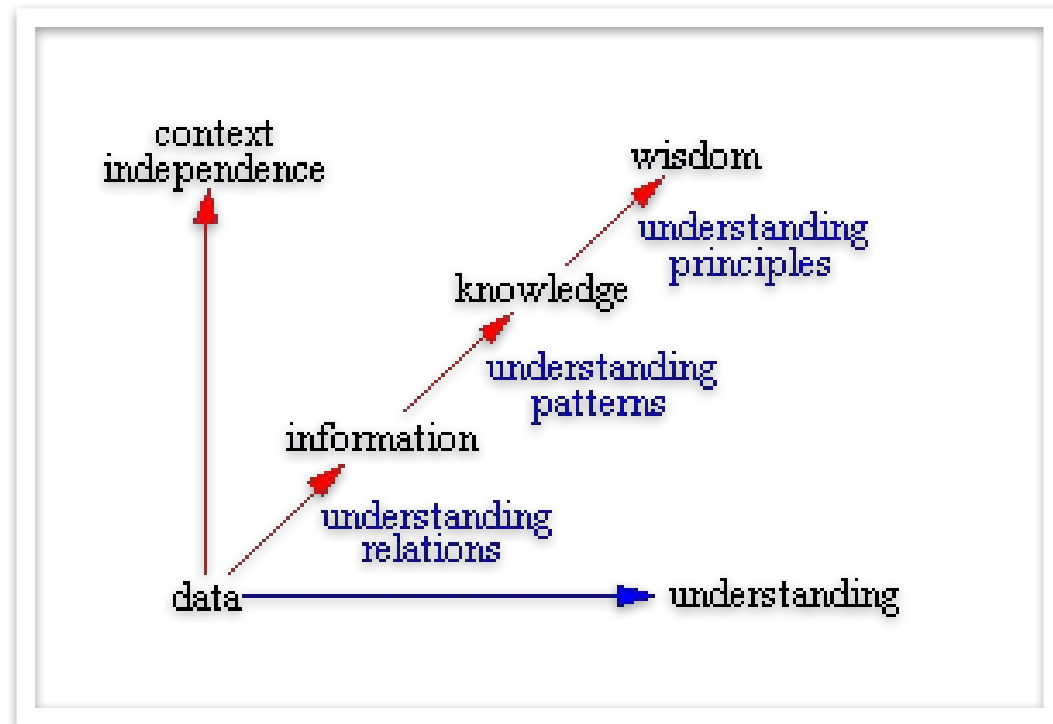
Appreciate the ability and skills to identify their information needs, source and to utilize the information efficiently and effectively



The MIS satisfies the diverse needs through a variety of systems such as Query Systems, Analysis Systems, Modeling Systems and Decision Support Systems the MIS helps in Strategic Planning, Management Control, Operational Control and Transaction Processing.

Knowledge Management

- **Definition:** Knowledge Management involves the acquisition, storage, retrieval, application, generation, and review of the knowledge assets of an organization in controlled way.



Dimensions of Knowledge

1. Tacit knowledge : is that stored in the brain of a person.

- Tacit knowledge is **personal**.
- It is accumulated through study and experience.
- It is developed through the process of interaction with other people.
- grows through the practice of **trial and error** and the **experience of success and failure**.

Disadvantage

- ❖ It is difficult to formalize, record, or articulate.
- ❖ It includes subjective insights, intuitions and estimations.
- ❖ As intuitive knowledge, it is **difficult to communicate and articulate**

NB. Tacit knowledge can be shared and communicated through various activities and mechanisms include conversations, **workshops, on-the-job training and the like**. Mechanisms include, among others, the use of information technology tools such as email, groupware, instant messaging and related technologies.

Dimensions of Knowledge

2. Explicit knowledge: is that **contained in documents** or other forms of storage other than the human.

- ❖ It is stored **in documents, databases, websites, emails and the like.**
- ❖ It is knowledge that can be **readily made available to others** and transmitted or shared in the form of systematic and formal languages
- These include knowledge assets such **as reports, memos, business plans, drawings, patents, trademarks, customer lists, methodologies, and the like**

Interaction between types of knowledge

- ❖ **Personal knowledge** can become **organizational knowledge** through the dynamic interaction between tacit knowledge and explicit knowledge.
- ❖ There are **four modes of knowledge creation** in an organization.
 1. Socialization,
 2. Externalization,
 3. Combination and
 4. Internalization

Interaction between types of knowledge

1. Socialization is a process of creating common tacit knowledge through shared experiences.

➤ The tacit knowledge of one person is shared and transmitted to another person and it becomes part of the other person's tacit knowledge.

2. Externalization is a process of articulating tacit knowledge into such explicit knowledge as concepts and/or diagrams.

➤ A good example of externalization is the process of creating a new product concept or developing a new production process.

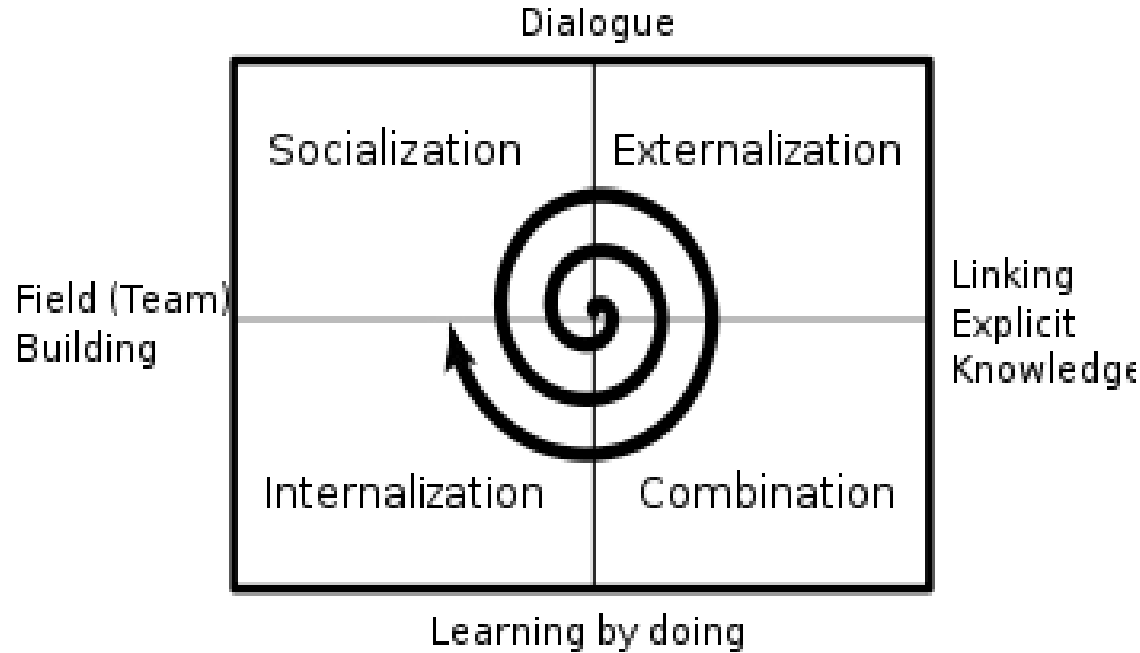
3. Combination: Is the process of combining or reconfiguring bodies of existing explicit knowledge in order to generate new explicit knowledge, by addition.

➤ It is the most common process in formal education.

Interaction between types of knowledge

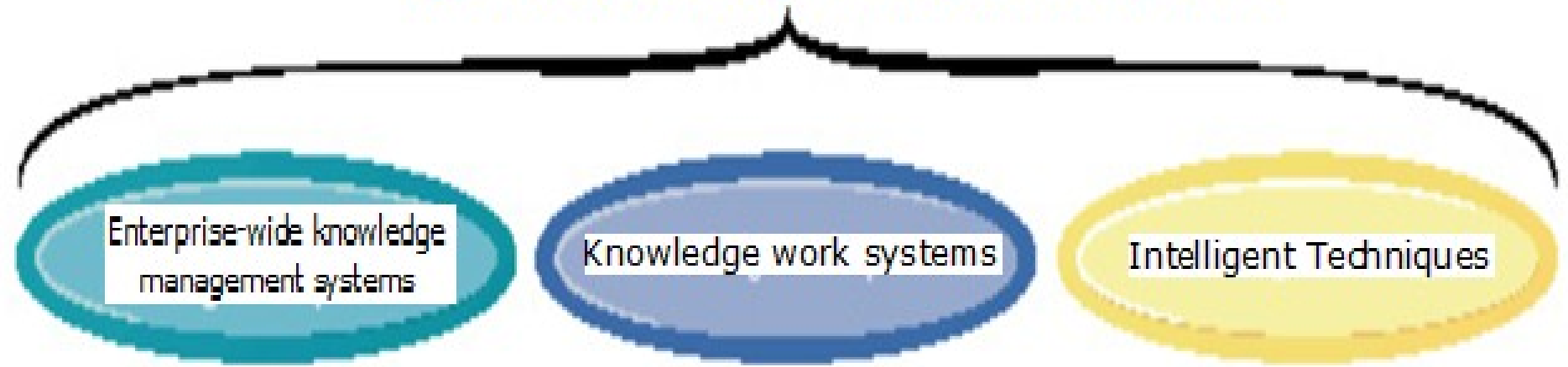
4. **Internalization** is a process of embodying explicit knowledge into tacit knowledge or an individual's know-how or operational knowledge.
- Explicit knowledge that is available as text, sound, or video facilitates the internalization process.
 - The instructions are learned and become part of the person's tacit knowledge.

Interaction between types of knowledge



Therefore, knowledge management is the conversion of tacit knowledge into explicit knowledge and sharing it within the organization.

Types of Knowledge Management Systems



General purpose, integrated, firm-wide efforts to collect, store, disseminate, and use digital content and knowledge

Structured knowledge systems
Semistructured knowledge systems
Knowledge network systems

Specialized workstations and systems that enable scientists, engineers, and other knowledge workers to create and discover new knowledge

Computer-aided design (CAD)
3D Visualization
Virtual reality
Investment workstations

Tools for discovering patterns and applying knowledge to discrete decisions and knowledge domains

Data mining
Neural networks
Expert systems
Case-based reasoning
Fuzzy logic
Genetic algorithms
Intelligent agents

Types of knowledge management systems

- **Enterprise**-wide knowledge management systems are general-purpose firm wide efforts to collect, store, distribute, and apply digital content and knowledge.

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Elements of Knowledge Management

- ❖ Complete knowledge management system must contain four elements.
 1. Knowledge creation and capture,
 2. Knowledge sharing and enrichment,
 3. Information storage and retrieval, and
 4. Knowledge dissemination

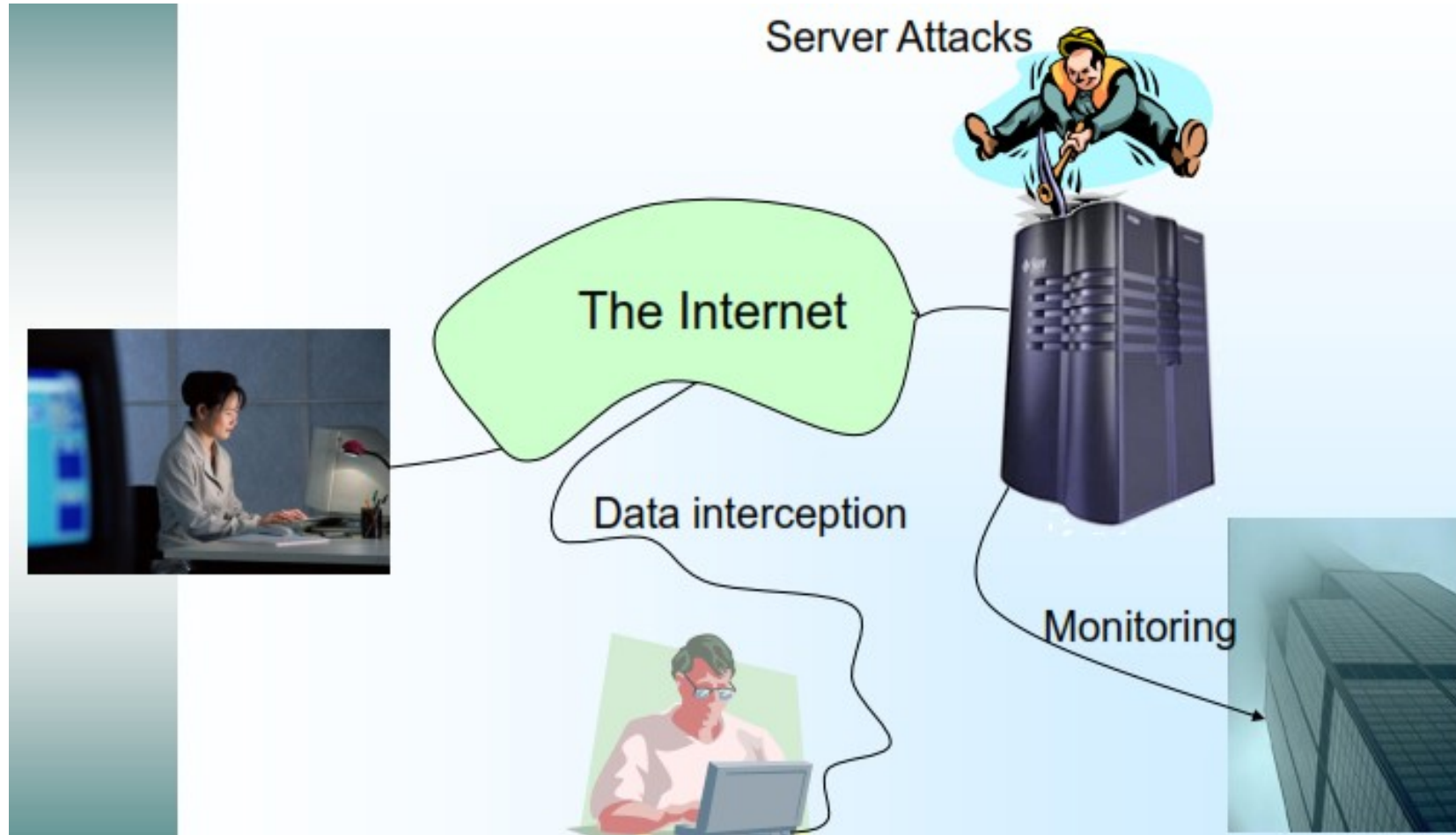
Self Assessment Questions

1. Distinguish between data, knowledge, and wisdom and between tacit knowledge and explicit knowledge.
2. Describe the stages in the knowledge management value chain.
3. Define and describe the various types of enterprise-wide knowledge management systems and explain how they provide value for businesses.
4. Define knowledge work systems and describe the generic requirements of knowledge work systems.

SECURITY AND ETHICAL ISSUES

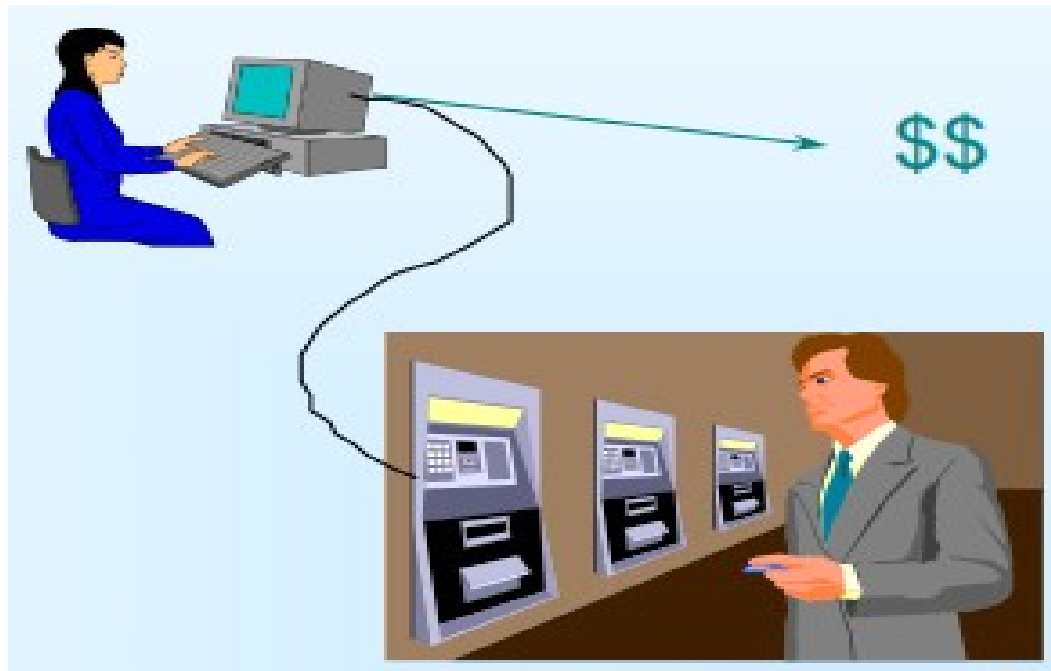
- ❖ The threats against, and defenses needed for the performance and security of business information systems, as well as the ethical implications and societal impacts of information technology.
- ❖ The use of information technology in business has major impacts on society, and thus raises serious ethical issues in the areas such as:
 - **Crime** (Hacking, Cyber Theft, Unauthorized use at work, Software Piracy, Computer viruses and worms)
 - **Privacy** (job stress, damaged arm and neck muscles, eye strain, radiation exposure, and even death by cause of computer –related job stress.
 - **Employment**
 - **Health**
 - **Working Conditions**
 - **NB.** Students should realize that information technology could **have a beneficial effect as well as a negative effect** in each of the areas listed above.

1.Computer Security



Security Categories

- Physical attack & disasters
- Backup--off-site
- Cold/Shell site
- Hot site
- Disaster tests



- ✓ Unauthorized modification
- ✓ Unauthorized disclosure
- ✓ Unauthorized withholding
- ✓ Rejection of Service



Figure A strategic opportunities matrix helps to evaluate the strategic risk/payoff potential of proposed business/IT opportunities

Tools of Security Management

- The goal of security management is the **accuracy, integrity, and safety** of an information system process and resources.
- Thus, the effecting security management can **minimize error, frauds, and losses** in the information systems that interconnect today's companies and their customers, suppliers, and their stock holders

Some important security defenses

- ✓ Encryption
- ✓ Firewalls
- ✓ Denial of services defenses
- ✓ E-Mail monitoring
- ✓ Virus defenses
- ✓ **Security Codes**
- ✓ **Backup Files**
- ✓ **Security Monitors**
- ✓ **Computer Failure control**

Self Assessment Questions

1. List and describe the most common threats against contemporary information systems.
 2. Define malware and distinguish among a virus, a worm, and a Trojan horse.
 3. Define a hacker and explain how hackers create security problems and damage systems.
 4. Explain how software defects affect system reliability and security.
 5. What is the business value of security and control?
 6. Distinguish between fault-tolerant and high availability computing, and between disaster recovery planning and business continuity planning.
 7. Identify and describe the security problems posed by cloud computing.
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Having Nice Time!!